

# A Review of the Filariasis Control Programme in Tahiti from November 1967 to January 1968

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*District filariasis surveys in Tahiti in the years after the Second World War yielded the highest microfilaria rates in the South Pacific area, ranging from 25 % to 44 %; the mean elephantiasis rate was 5 % and microfilaria densities (MfD<sub>50</sub>) ranged from 18 to 31. A co-operative filariasis research programme was begun and a control programme was inaugurated in 1953, using diethylcarbamazine therapy supported by mosquito larvae control measures. By 1959 fifteen districts had received mass treatment. Re-examination after 12 months showed that the mean microfilaria rate had dropped from 31 % to 3 % and the MfD<sub>50</sub> from 23 to 4. Subsequently, only positives were re-treated and by 1964 the microfilaria rate had risen to 6.8 %. The increase caused concern and the significance of this reservoir of infection was investigated.*

*Administration of diethylcarbamazine in periodic mass treatments only in American Samoa began in 1963 and was accompanied by a drop in the microfilaria rate to less than 1 %, leading to interruption of transmission in some areas.*

This paper reviews a programme for the control of subperiodic filariasis based primarily on the therapeutic use of diethylcarbamazine, with the elimination of mosquito breeding places within 100 m of each household as a secondary measure. The study was undertaken to answer the question why, during some 12 years of systematic control in Tahiti, the microfilaria (mf) rate has persisted at about 3-5 %.

The review is limited, in general, to repeat analyses of the early data obtained in 3 representative rural districts: Vairao, Mataiea, and Tautira (Fig. 1). Data from these districts had not previously been documented as recommended by the WHO Expert Committee on Filariasis (1967) and it was decided that they should be re-organized in accordance with the recommendations made in that report.

(1) Vairao District, extending from kilometre 2 to kilometre 14 in the peninsula known as Little Tahiti, is composed of 2 subdistricts and had a population of 825, all of whom were examined in the first blood survey in 1950. This district was

selected as the control area and diethylcarbamazine treatment was withheld.

(2) Two additional districts, Mataiea and Tautira, with a combined population of 1200 persons, were also selected for the review. They were among the 15 rural districts included in the mass treatment programme that began in 1953. Mataiea was observed from 1949 to 1966 and Tautira from 1950 to 1968. The latter was the first district to receive mass treatment in 1953 and, under the author's supervision, was again subjected to complete follow-up blood surveys and intensive mosquito surveys in 1958 and 1968. The microfilaria rates and densities from these three districts are compared at various intervals.

## PROCEDURES

Special methods used in the programme are as follows.

### *Blood surveys*

Initial blood surveys before the administration of diethylcarbamazine were based on the examination of a thick blood film (20 mm<sup>3</sup>) obtained by finger puncture and stained with Giemsa. Final follow-up surveys after treatment were based on the examination of an average of two or three 20-mm<sup>3</sup> blood films.

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Table 1. Infective rates of filariasis at Vairao (entire district), Tahiti, for 1950, 1953, and 1954-56

| Microfilarial count | 1950         |                   |                       | 1953         |                   |                       | 1954-6       |                   |                       |
|---------------------|--------------|-------------------|-----------------------|--------------|-------------------|-----------------------|--------------|-------------------|-----------------------|
|                     | No. of cases | Cumulative number | Cumulative percentage | No. of cases | Cumulative number | Cumulative percentage | No. of cases | Cumulative number | Cumulative percentage |
| 1                   | 14           | 14                | 5.5                   | 8            | 8                 | 4.5                   | 16           | 16                | 5.8                   |
| 2                   | 10           | 24                | 9.4                   | 12           | 20                | 11.3                  | 6            | 22                | 7.9                   |
| 3                   | 8            | 32                | 12.5                  | 8            | 28                | 15.8                  | 17           | 39                | 14.1                  |
| 4                   | 8            | 40                | 15.7                  | 5            | 33                | 18.6                  | 7            | 46                | 16.6                  |
| 5                   | 7            | 47                | 18.4                  | 3            | 36                | 20.3                  | 13           | 59                | 21.3                  |
| 6                   | 4            | 51                | 20.0                  | 4            | 40                | 22.6                  | 8            | 67                | 24.2                  |
| 7                   | 15           | 66                | 25.9                  | 4            | 44                | 24.9                  | 5            | 72                | 26.0                  |
| 8                   | 3            | 69                | 27.0                  | 2            | 46                | 26.0                  | 3            | 75                | 27.1                  |
| 9                   | 3            | 72                | 28.2                  | 7            | 53                | 30.0                  | 2            | 77                | 27.8                  |
| 10                  | 4            | 76                | 29.8                  | 3            | 56                | 31.6                  | 5            | 82                | 29.6                  |
| 11-20               | 36           | 112               | 43.9                  | 23           | 79                | 44.6                  | 41           | 123               | 44.4                  |
| 21-30               | 10           | 122               | 47.8                  | 10           | 89                | 50.3                  | 22           | 145               | 52.3                  |
| 31-40               | 12           | 134               | 52.5                  | 17           | 106               | 59.9                  | 20           | 165               | 59.5                  |
| 41-50               | 6            | 150               | 54.9                  | 6            | 112               | 63.3                  | 15           | 180               | 65.0                  |
| 51-60               | 12           | 152               | 59.6                  | 9            | 121               | 68.4                  | 11           | 191               | 69.0                  |
| 61-70               | 6            | 158               | 62.0                  | 6            | 127               | 71.8                  | 3            | 197               | 71.1                  |
| 71-80               | 10           | 168               | 65.9                  | 3            | 130               | 73.4                  | 9            | 206               | 74.3                  |
| 81-90               | 6            | 174               | 68.2                  | 7            | 137               | 77.4                  | 17           | 223               | 81.0                  |
| 91-100              | 6            | 180               | 70.6                  | 3            | 140               | 79.1                  | 3            | 226               | 81.7                  |
| 101-200             | 32           | 212               | 83.1                  | 18           | 158               | 89.3                  | 21           | 247               | 89.2                  |
| 201-300             | 15           | 227               | 89.0                  | 6            | 164               | 92.7                  | 16           | 263               | 95.0                  |
| 301-400             | 14           | 241               | 94.5                  | 6            | 170               | 96.0                  | 7            | 270               | 97.5                  |
| 401-500             | 4            | 245               | 96.1                  | 3            | 173               | 97.7                  | 1            | 271               | 97.8                  |
| 501-600             | 5            | 250               | 98.0                  | 3            | 176               | 99.4                  | 4            | 275               | 99.2                  |
| 601-700             | 1            | 251               | 98.4                  | 0            |                   |                       | 0            |                   |                       |
| 701-800             | 1            | 252               | 98.8                  | 0            |                   |                       | 1            | 276               | 99.6                  |
| 801-900             | 0            |                   |                       | 0            |                   |                       | 0            |                   |                       |
| 901-1 000           | 1            | 253               | 99.2                  | 0            |                   |                       | 0            |                   |                       |
| >1 000              | 2            | 255               | 100.0                 | 1            | 177               | 100.0                 | 1            | 277               | 100.0                 |

*Control area*

Early in the filariasis programme in Tahiti, the entire district of Vairao mentioned under (1) above was given a first blood survey in 1950 and a second in 1953; kilometres 2-8 were surveyed in 1954 and the remainder of the area was surveyed in 1956, before diethylcarbamazine treatment (see Kessel, 1957, Tables II and VIII, and Table 1 and Fig. 2 of this

paper). Altogether, 13 intensive mosquito surveys were performed, as described by Bonnet et al. (1956), between 1953 and 1956 and the results were summarized by Kessel (1967). The infective-stage larval rate of *W. bancrofti* in 7433 *Aedes polynesiensis* dissected during the 3 years was 3.1%, the lowest rate being 1.3% and the highest 7.6%.

When the results of simultaneous standardized

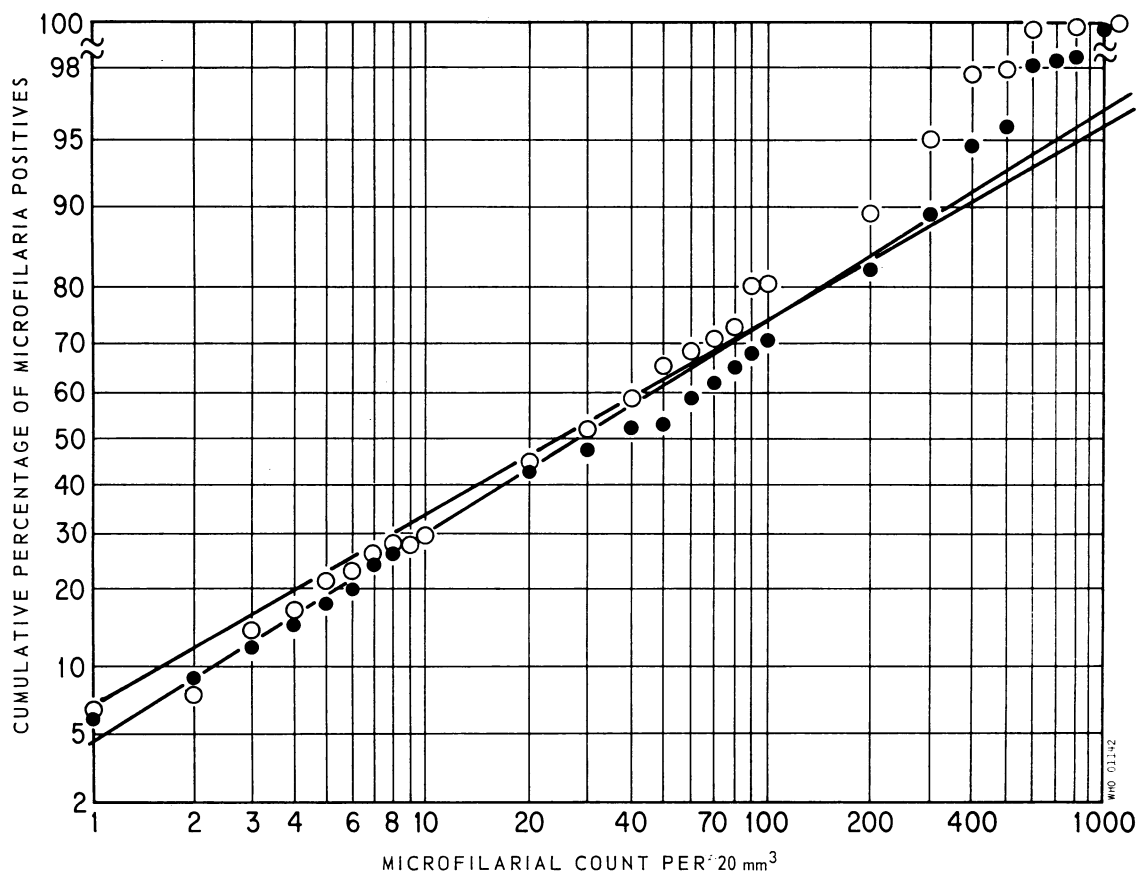


Fig. 2. Cumulative percentage distribution of microfilaria positives and microfilarial counts for Vairao (entire district). Solid circles: 1950; open circles: 1956.

blood and mosquito surveys are compared, close positive correlations are found between the mf rate and the  $MfD_{50}$  in man, and the infective stage larval rates in dissected mosquitos. This correlation demonstrates the value of using both types of survey to monitor the progress of a control campaign in a given area.

#### *Drug control*

The successful control and treatment of clinical filariasis with drugs was, in comparison with that of other helminthic diseases, retarded prior to the Second World War because antimonial and arsenical compounds, especially arsenoxides, could not be used in the treatment of human filariasis although these substances gave promising results in parasitic infections of animals. The discovery by Hewitt et al. (1947a, 1947b, 1947c) of the value of piperazine

derivatives in filaria infections in the cotton rat and the dog led to the proposal to use diethylcarbamazine to treat human filariasis. The first use of this compound in the South Pacific area was reported by Galliard et al. (1949) and it has subsequently been used in many areas of the Pacific (see Beye et al., 1952, 1953; Bonnet et al., 1956; Burnett, 1964; Cifferi & Kessel, 1967; Cifferi et al., 1969; Fukushima, 1967; Kessel, 1957; Kessel & Massal, 1962; Kessel et al., 1970; Laigret et al., 1966; McCarthy & Carter, 1964; McCarthy & Fitzgerald, 1955; Marshall & Yasukawa, 1966; Sasa, 1963; Sasa et al., 1960; Sato et al., 1960; Symes, 1960; also Thieme et al., 1968, unpublished report to WHO).

#### *Regimens for diethylcarbamazine treatment in Tahiti*

When diethylcarbamazine was first tested by Beye et al. (1952) dosages of 2 mg per kg of body

weight 3 times a day for 7 days were given. At the end of the year the microfilaria rate stood at 19%, having been reduced from 41%. The average number of microfilariae per 20 mm<sup>3</sup> of blood in the whole population had dropped from 56 to 1.6. The reductions were, however, temporary and insufficient to warrant the use of such a schedule for control programmes. After comparing several additional regimens suggested by Beye et al. (1953) and Thooris et al. (1956), a dosage rate of 6 mg/kg to be given once a month for 12 months was selected as a minimum level of treatment but it was decided to recommend treatment for 24 months as the standard mass-treatment regimen for the control of filariasis.

On the completion of 24 doses, treatment was discontinued and the first follow-up blood survey was undertaken 1 year later. All persons found to be positive for microfilariae during the first follow-up were re-treated monthly for a further year; all positives detected at each subsequent annual survey after the first were given the standard repeat regimen of 72 mg/kg of diethylcarbamazine, although some variations in schedules were recommended in 1960. In 1967, periodic mass treatment was introduced into Moorea by the Director and staff of the Institut de Recherches Médicales.

Diethylcarbamazine appears to be the only safe drug that has yet been used in extensive filariasis control trials, although the WHO Expert Committee on Filariasis (1962, 1967) has urged that new drugs should be screened and tested in both animals and man. For recent developments in laboratory animals, see Ash et al. (1970a, 1970b) and Schacher et al. (1969, 1970).

Insufficient information is available concerning the pharmacological action of diethylcarbamazine, although it has been repeatedly demonstrated that the best results have been observed in association with well-planned and strictly supervised control programmes, employing adequate dosages of diethylcarbamazine. Periodic follow-up surveys from different areas show that marked reductions occurred in mf rates, MfD<sub>50</sub> values, infective-stage larval rates of the parasite mosquitos, and clinical filariasis rates. The renewed interest and activity in these fields is therefore timely.

## RESULTS AND DISCUSSION

### *Non-treated areas*

Periodic blood surveys in the control area of Vairao made in 1950, 1953, and 1954–56 are illus-

trated in Table 1 and Fig. 2. The mf rate in 1950 was 30.9% and the MfD<sub>50</sub> was 28; in 1956, the mf rate was 31.7% and the MfD<sub>50</sub> was 26, showing no significant difference even at the  $P = 0.05$  level. Without the use of diethylcarbamazine, these studies thus indicate almost identical mf rates and MfD<sub>50</sub> values for 6 years.

Ciferri et al. (1969) and Kessel et al. (1970) compared surveys made in American Samoa in 1945 by Murray (1948) and by Jachowski in 1948 (Jachowski, personal communication) with later surveys carried out in 1962–63 in the same villages where no diethylcarbamazine had been administered in control programmes. They were able to show that similar mf rates and MfD<sub>50</sub> values had been maintained in 5 villages for some 15–20 years. Both studies showed that when diethylcarbamazine was not administered there was little change in mf rates and in the MfD<sub>50</sub>. Subsequent administration of diethylcarbamazine in Vairao, beginning in 1956 with the schedule used in Tahiti, resulted in an mf rate of 3% and an MfD<sub>50</sub> of 3.0 after 4 years. Following the administration of diethylcarbamazine in the 5 villages in American Samoa referred to above, comparable reductions in mf rates and MfD<sub>50</sub> values were found.

### *Treated areas*

Filariasis control by mass treatment with diethylcarbamazine was begun in Tahiti in 1953 and by 1957, when the first 15 districts had received mass treatment for 3–4 years and follow-up blood and mosquito surveys also had been completed, the mf rate had dropped from 31% to 3.0% and the infective-stage larval rate had dropped to less than 0.5%. These results were most encouraging. Soon after, however, with positives only being treated, the mf rate began to rise and by 1960 the over-all rate had reached 6%, with a minimum of 3.8% and a maximum of 10.3%.

Recently, a repeat analysis of the data from Mataiea and Tautira, 2 of the original 15 representative rural districts, was made by means of the frequency distribution formula suggested by the WHO Expert Committee on Filariasis (1967).

### *Mataiea and Tautira*

The filariasis situation before mass treatment began, and the results 3–4 years after the close of mass treatment and again after another 8–10 years, are shown in Table 2. At first, there was a marked fall in the mf rate from 39% in Mataiea and 27% in

Table 2. Results of long-term studies with diethylcarbamazine mass treatment in Tahiti<sup>a</sup>

| District                                     | Years of surveys | No. of persons examined | Mf rate (%) | Mf D <sub>50</sub> |
|----------------------------------------------|------------------|-------------------------|-------------|--------------------|
| Before mass treatment                        |                  |                         |             |                    |
| Mataiea                                      | 1949             | 619                     | 39          | 23                 |
| Tautira                                      | 1950             | 538                     | 27          | 18                 |
| Third or fourth years after mass treatment   |                  |                         |             |                    |
| Mataiea                                      | 1958             | 713                     | 4.9         | 4.5                |
| Tautira                                      | 1958             | 570                     | 3.3         | 4.0                |
| Tenth or eleventh years after mass treatment |                  |                         |             |                    |
| Mataiea                                      | 1966             | 976                     | 5.2         | 3.5                |
| Tautira                                      | 1966             | 794                     | 4.0         | 3.5                |

<sup>a</sup> Mass treatment began in 1953 in Tautira and in 1954 in Mataiea. The regimens in both areas were 6 mg/kg monthly for 24 months (total, 144 mg/kg); then positives only re-treated annually.

Tautira before treatment to 1% at the close of mass treatment. By 1958, the mf rate had increased to 4.9% in Mataiea and to 3.3% in Tautira. Even so, the differences between pre-treatment and post-treatment mf rates were significant at the  $P < 0.0001$  level.

Similar changes were noted in the MfD<sub>50</sub> values, which dropped from 23 in Mataiea and 18 in Tautira before treatment to a low value of 1 at the close of treatment. By 1958, however, the third year after the close of mass treatment, the MfD<sub>50</sub> had increased to 4.5 in Mataiea and to 4.0 in Tautira. Some 10 years later, after only those found to be microfilaria-positive in annual blood surveys had been re-treated, the mf rate stood at 5.2% in Mataiea and at 4.0% in Tautira, and the MfD<sub>50</sub> value was 3.5 in both districts. The latter increased to 4.1 in 1968 in Tautira, as shown in Tables 4 and 5 and Fig. 3.

Table 3 shows microfilarial counts, the number of cases, and the cumulative percentages of microfilaria-positive persons in the district of Mataiea for the years 1949 (before diethylcarbamazine treatment), 1958 (4 years after mass treatment began), and 1966 (10 years after the institution of the programme of re-treating only microfilaria-positive persons). The MfD<sub>50</sub> value is determined from

such cumulative percentage tables, being the point where the regression line meets the 50-percentile line.

Table 4 and Fig. 3 show similar data for the district of Tautira in 1950 (before diethylcarbamazine treatment), in 1958 (5 years after mass treatment began), and again 10 years later in 1968. The MfD<sub>50</sub> is shown to be 18 in 1950, 4 in 1958, and 4.1 in 1968.

Table 5 summarizes a correlation of 5 blood surveys and 5 intensive mosquito surveys in Tautira, the first of the 15 districts to receive mass treatment with diethylcarbamazine. The results are similar to those from Mataiea and other districts studied at the same time and display a characteristic pattern.

When comparing the results in Tahiti with those of other studies, e.g., those of Sato et al. (1960), Sasa (1963, 1966), and Fukushima (1967) in Japan, and those of Ciferri & Kessel (1967), Ciferri et al. (1969), and Kessel et al. (1971) in American Samoa, it appears, within limits, that the success of diethylcarbamazine treatment for filariasis is related to the ratio between the microfilaria load and the dosage of diethylcarbamazine. The thoroughness with which the programme is administered also is of major importance.

Factors involved in the host-parasite relationship that influence variability in the success of a control programme are:

- (1) the mf rate in man;
- (2) the MfD<sub>50</sub>, or density of microfilariae per unit volume of blood, usually reported per 20 mm<sup>3</sup>; and
- (3) the population density of known mosquito vectors.

Periodic follow-up blood surveys help to determine the status or progress of a control programme and may be evaluated with the aid of either prevalence rates or incidence rates. Surveys are influenced by the following factors:

- (1) The number and origin of untreated immigrants moving into a treated area. These people should be checked immediately they arrive in the new community and should be treated at once. One reason for a high untreated immigrant rate in Tahiti during the period under review was the influx of workers from outer French Polynesia to assist with preparations for atomic studies. Many came from islands where filariasis control was not enforced.

Table 3. Infective rates of filariasis at Mataiea (entire district), Tahiti, for 3 periods before and after mass treatment with diethylcarbamazine

| Microfilarial count | Before mass treatment, 1949 |                   |                       | After mass treatment, 1959 |                   |                       | After mass treatment, 1968 |                   |                       |
|---------------------|-----------------------------|-------------------|-----------------------|----------------------------|-------------------|-----------------------|----------------------------|-------------------|-----------------------|
|                     | No. of cases                | Cumulative number | Cumulative percentage | No. of cases               | Cumulative number | Cumulative percentage | No. of cases               | Cumulative number | Cumulative percentage |
| 1                   | 10                          | 10                | 7.4                   | 8                          | 8                 | 19.0                  | 8                          | 8                 | 25.0                  |
| 2                   | 5                           | 15                | 11.0                  | 1                          | 9                 | 21.4                  | 4                          | 12                | 37.5                  |
| 3                   | 4                           | 19                | 14.0                  | 4                          | 13                | 31.0                  | 3                          | 15                | 46.9                  |
| 4                   | 4                           | 23                | 17.0                  | 5                          | 18                | 42.9                  | 1                          | 16                | 50.0                  |
| 5                   | 3                           | 26                | 19.1                  | 1                          | 19                | 45.2                  | 2                          | 18                | 56.3                  |
| 6                   | 1                           | 27                | 19.9                  | 1                          | 20                | 47.6                  |                            |                   |                       |
| 7                   | 2                           | 29                | 21.3                  | 2                          | 22                | 52.4                  |                            |                   |                       |
| 8                   | 1                           | 30                | 22.1                  | 0                          |                   |                       |                            |                   |                       |
| 9                   | 2                           | 32                | 23.5                  | 1                          | 23                | 54.8                  | 1                          | 19                | 59.4                  |
| 10                  | 0                           | 32                | 23.5                  | 0                          |                   |                       | 0                          |                   |                       |
| 11–20               | 20                          | 52                | 38.2                  | 10                         | 33                | 78.6                  | 7                          | 26                | 81.3                  |
| 21–30               | 18                          | 70                | 51.5                  | 6                          | 39                | 92.9                  | 4                          | 30                | 93.8                  |
| 31–40               | 12                          | 82                | 60.3                  | 0                          |                   |                       | 0                          |                   |                       |
| 41–50               | 4                           | 86                | 63.2                  | 2                          | 41                | 97.6                  | 0                          |                   |                       |
| 51–60               | 8                           | 94                | 69.1                  |                            |                   |                       | 0                          |                   |                       |
| 61–70               | 5                           | 99                | 72.8                  |                            |                   |                       | 1                          | 31                | 96.9                  |
| 71–80               | 7                           | 106               | 77.9                  |                            |                   |                       | 0                          |                   |                       |
| 81–90               | 5                           | 111               | 81.6                  |                            |                   |                       | 0                          |                   |                       |
| 91–100              | 1                           | 112               | 82.4                  | 1                          | 42                | 100.0                 | 0                          |                   |                       |
| 101–200             | 12                          | 124               | 91.2                  |                            |                   |                       | 1                          | 32                | 100.0                 |
| 201–300             | 7                           | 131               | 96.3                  |                            |                   |                       |                            |                   |                       |
| 301–400             | 1                           | 132               | 97.1                  |                            |                   |                       |                            |                   |                       |
| 401–500             | 2                           | 134               | 98.5                  |                            |                   |                       |                            |                   |                       |
| 501–600             |                             |                   |                       |                            |                   |                       |                            |                   |                       |
| 601–700             | 2                           | 136               | 100.0                 |                            |                   |                       |                            |                   |                       |

(2) The number of persons who refuse treatment or who are excused in a control programme.

(3) The persistent microfilaraemias at each follow-up survey.

(4) New infections or new microfilaraemias, including reinfections.

(5) Recurrent microfilaraemias, best evaluated by incidence rates (Mahoney & Kessel, unpublished data).

The review of the data from the annual follow-up blood surveys for 1966 in Mataiea and Tautira

involved 1770 persons and revealed the following positives; 23 new immigrants, 15 new infections, 10 recurrences, and 9 refused or excused, making a total of 57 or 3.2%. The presence of such continuous microfilaraemia indicates the presence of a smouldering reservoir of filariasis in man although it is often so reduced that clinical filariasis is no longer considered to be a major public health problem. March et al. (1960) first noted in Tahiti, 7 years after the beginning of mass treatment with diethylcarbamazine that the elephantiasis and hydrocele rates were falling, while Outin & Bambridge (personal

Table 4. Infective rates of filariasis at Tautira (entire district), Tahiti, for 1950, 1958, and 1968

| Microfilarial count | 1950         |                   |                       | 1958         |                   |                       | 1968         |                   |                       |
|---------------------|--------------|-------------------|-----------------------|--------------|-------------------|-----------------------|--------------|-------------------|-----------------------|
|                     | No. of cases | Cumulative number | Cumulative percentage | No. of cases | Cumulative number | Cumulative percentage | No. of cases | Cumulative number | Cumulative percentage |
| 1                   | 21           | 21                | 13.5                  | 4            | 4                 | 21.0                  | 6            | 6                 | 18.1                  |
| 2                   | 12           | 33                | 21.2                  | 1            | 5                 | 26.3                  | 4            | 10                | 30.3                  |
| 3                   | 1            | 34                | 21.8                  | 3            | 8                 | 42.1                  | 3            | 13                | 39.3                  |
| 4                   | 1            | 35                | 22.4                  | 2            | 10                | 52.6                  | 3            | 16                | 48.4                  |
| 5                   | 4            | 39                | 25.0                  | 0            |                   |                       | 0            |                   |                       |
| 6                   | 0            |                   |                       | 0            |                   |                       | 1            | 17                | 51.5                  |
| 7                   | 7            | 46                | 29.5                  | 1            | 11                | 57.9                  | 0            |                   |                       |
| 8                   | 3            | 49                | 31.4                  | 0            |                   |                       | 3            | 20                | 60.6                  |
| 9                   | 4            | 53                | 34.0                  | 1            | 12                | 63.1                  | 2            | 22                | 66.6                  |
| 10                  | 0            |                   |                       | 1            | 13                | 68.4                  | 2            | 24                | 72.7                  |
| 11- 20              | 16           | 69                | 44.2                  | 4            | 17                | 89.5                  | 1            | 25                | 75.7                  |
| 21- 30              | 14           | 83                | 53.2                  | 1            | 18                | 94.7                  | 6            | 31                | 93.9                  |
| 31- 40              | 12           | 95                | 60.9                  | 0            |                   |                       | 1            | 32                | 96.9                  |
| 41- 50              | 6            | 101               | 64.7                  | 1            | 19                | 100.0                 | 1            | 33                | 100.0                 |
| 51- 60              | 10           | 111               | 71.2                  |              |                   |                       |              |                   |                       |
| 61- 70              | 9            | 120               | 76.9                  |              |                   |                       |              |                   |                       |
| 71- 80              | 5            | 125               | 80.1                  |              |                   |                       |              |                   |                       |
| 81- 90              | 3            | 128               | 82.1                  |              |                   |                       |              |                   |                       |
| 91-100              | 2            | 130               | 83.3                  |              |                   |                       |              |                   |                       |
| 101-200             | 17           | 147               | 94.2                  |              |                   |                       |              |                   |                       |
| 201-300             | 5            | 152               | 97.4                  |              |                   |                       |              |                   |                       |
| 301-400             | 2            | 154               | 98.7                  |              |                   |                       |              |                   |                       |
| 401-500             | 1            | 155               | 99.4                  |              |                   |                       |              |                   |                       |
| 501-600             | 1            | 156               | 100.0                 |              |                   |                       |              |                   |                       |

Table 5. Correlations of blood surveys and intensive mosquito surveys in Tautira, Tahiti

| Mass treatment with diethylcarbamazine | Blood surveys for microfilariae in man |          |                   | Mosquito surveys for infective-stage microfilariae |                     |
|----------------------------------------|----------------------------------------|----------|-------------------|----------------------------------------------------|---------------------|
|                                        | No. positive/no. examined              | Rate (%) | MfD <sub>50</sub> | Mosquitos positive/mosquitos dissected             | Percentage positive |
| before treatment (1950)                | 156/583                                | 26.6     | 18.0              | 147/4733                                           | 3.1 <sup>a</sup>    |
| after treatment (year; no. of years)   |                                        |          |                   |                                                    |                     |
| (1953; 1)                              | 35/576                                 | 6.1      | 6.0               | 6/1495                                             | 0.4                 |
| (1958; 5)                              | 19/570                                 | 3.3      | 4.0               | 0/116                                              | 0.0                 |
| (1966; 13)                             | 32/794                                 | 4.0      | 3.5               | 0/80                                               | 0.0                 |
| (1968; 15)                             | 33/739                                 | 4.5      | 4.1               | 1/299                                              | 0.3                 |

<sup>a</sup> Control district of Vairao, 1953-56 substituted since method for mosquito collections was not yet standardized in 1950.



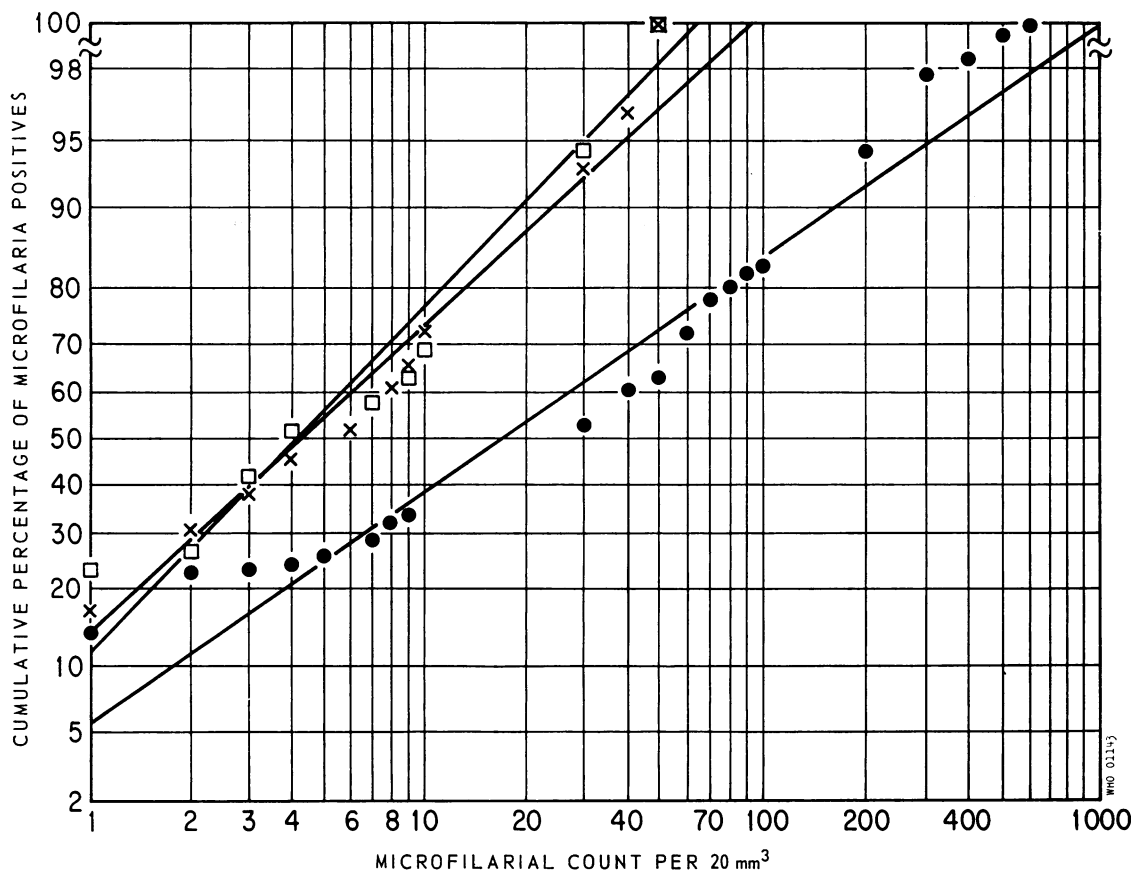


Fig. 3. Cumulative percentage distribution of microfilaria positives and microfilarial counts for Tautira (entire district). Solid circles: 1950; squares: 1958; crosses: 1968.

communication) have found only 2 new autochthonous cases of light elephantiasis in the last 10 years in the same rural districts where Beye et al. (1952) reported an elephantiasis rate of 5%.

#### *Reduction of filariasis in younger age groups*

Another comparison of special significance in evaluating a programme is the reduction in mf rates and in the  $MfD_{50}$  in younger age groups, e.g., 0–9 years—that is, among children born during the period after which control was routinely established but who themselves received no diethylcarbamazine. It is presumed that children are negative for filaria at birth and that those who subsequently show filarial infections have acquired them from infected mosquitos. Beye et al. (1953) have reported that before administration of diethylcarbamazine 205 children among 1452 examined were microfilaria-positive

(mf rate of 14.1%). Laigret et al. (1965) found only 18 children positive among 1486 in the 0–9 years age group examined in the same rural area (mf rate of 1.2%). The average microfilarial density had dropped from 32 to 5. This indicates a remarkable reduction in both mf rate and  $MfD_{50}$ . The corresponding reduction in the 0–4 years age group was from 4.8% to 0.4% during this period.

#### *Relation of microfilariae load to treatment with diethylcarbamazine*

The possible relationship of mf rate and  $MfD_{50}$  to clinical filariasis and to response to therapeutic measures and to mass treatment for control has often been discussed in the literature. Combined high microfilaria rates and densities are generally associated with high clinical rates, but high microfilaria rates may be reported with low clinical rates in

elephantiasis when the microfilaria densities are low, as shown by Pipken (1953) in Micronesia. There appears however to be a correlation between microfilaria densities and response to diethylcarbamazine therapy, i.e., carriers with a high  $MfD_{50}$  respond less readily than those with a low  $MfD_{50}$ , as shown by Ciferri & Kessel (1967).

In discussions between the author and Professor M. Sasa it was noted that both the mf rate and the  $MfD_{50}$  were much higher in certain areas (e.g., Tahiti, where demonstrated  $MfD_{50}$  values range from 20 to 30) than in others (e.g., Japan, where the highest  $MfD_{50}$  found was 16.6; see Sasa, 1966, Fig. 8). Subsequent reports of microfilaria rates and densities in American Samoa demonstrated intermediate levels. The levels in Japan, American Samoa, and Tahiti are as follows:

(1) Japan: low. The highest mf rate recorded before diethylcarbamazine treatment was recorded by Sasa (1966) as 15.4% and the highest  $MfD_{50}$  was 16.6.

(2) American Samoa: intermediate. The mf rate among all ages in villages before diethylcarbamazine treatment ranged from 15% to 25% and the  $MfD_{50}$  from 12 to 33 (Ciferri et al., 1969, Table 6).

(3) Tahiti: high. The mf rate in districts before diethylcarbamazine treatment ranged from 25% to 44% and the  $MfD_{50}$  from 18 to 35 (Beye et al., 1953).

#### *Periodic treatment of positives only compared with periodic mass treatment*

Many schedules have been recommended for the administration of diethylcarbamazine, the cost of the programme, the cost of the drug and the time involved in administering treatment being important considerations. The frequency of follow-up programmes has also to be considered. In Brazil, Rachou & Ferreira (1958) recommended, for reasons of economy, that mass treatment should be given to the whole population of an area at suitable recurrent intervals without annual follow-up microfilaria surveys.

In the programme that began in 15 districts in Tahiti in 1953 it was recommended that 6 mg of diethylcarbamazine per kg of body weight should be administered once a month for 24 months. Subsequently, annual follow-up blood surveys were undertaken and only the positives were re-treated. Current reports from Tahiti still show an over-all mf rate of about 4% and an  $MfD_{50}$  of approximately 4.

In American Samoa, periodic mass treatment with

a total regimen of 72 mg/kg was recommended every 2 years until treatment was no longer needed. Follow-up surveys some 18 months after each mass treatment showed that following the second round of treatment, the mf rate had dropped to less than 0.38% (Kessel, 1971). A third mass treatment will probably be necessary.

Fukushima (1967, Fig. 7), working in Kagoshima Prefecture, Japan, where there was an initial mf rate of 11.2% before diethylcarbamazine treatment and an  $MfD_{50}$  of 6, reported "eradication" of filariasis following 3 mass treatments between 1960 and 1965 with a total of 72 mg of diethylcarbamazine per kg of body weight in each mass treatment.

Kessel (1971) reported the first instance of interruption of transmission in American Samoa, which lasted at least 2 years in the village of Amouli. This village had shown a low pre-treatment mf rate of 15% and an  $MfD_{50}$  of 12. Two mass treatments of 72 mg of diethylcarbamazine per kg of body weight were given, the first in 1962-63 and the second in 1965-66.

It is of interest to note that the original microfilaria rates and densities were only slightly higher in Amouli, American Samoa, than in Kagoshima, Japan, and that both areas were given periodic mass treatment rather than treatment only of positives at annual follow-up blood surveys.

#### *Control compared with eradication*

Reports from Japan, American Samoa, and Tahiti have emphasized the problems encountered in three filariasis control programmes based primarily on mass treatment with diethylcarbamazine. Similar approaches have been used in different parts of the world. Some have been sufficiently detailed and continued for a sufficient length of time to make the comparisons valid, e.g., (1) microfilaria rates and densities found in follow-up blood surveys; (2) rates of infective-stage *Wuchereria* larvae found in vector mosquitos; and (3) reductions in clinical filariasis of such significance as to indicate that the disease is declining in importance as a major public health problem.

As yet there is no uniformity of opinion among investigators regarding the desirability of filariasis control compared with filariasis eradication. Possibly, the reduction of microfilaraemia to 4% or less and an  $MfD_{50}$  of about 4.0 is sufficient to keep clinical filariasis down to a level at which it is no longer an important public health problem. If this is in fact the case, then only control of the type

practised since 1953 in Tahiti is necessary. If, however, transmission can be interrupted in a few small areas (e.g., villages, districts, or small islands), as shown in Amouli, American Samoa, then it may be possible to expand such areas to involve whole districts, as reported by Fukushima (1967). Treatment of entire larger islands or countries also seems to be feasible.

When mf rates are 1% or less, the  $MfD_{50}$  is also 1.0 or less and when standard mosquito surveys in these same areas reveal the absence of infective-stage larvae in all mosquitos dissected, it would appear to be opportune to standardize procedures for initiating

an eradication programme. For either control or eradication of filariasis, agreement on definitions and terms to be used in discussions should be achieved and it is hoped that attempts to draw up a glossary on filariasis will be continued.

In planning an eradication programme, the usual problems arise; they include questions of the costs and importance of filariasis in comparison with those of other disease problems in the area, and the amount of interest shown by the administration of the area. Above all, an adequate working budget and a well-trained field director are essential to the programme.

## RÉSUMÉ

### PROGRAMME DE LUTTE CONTRE LA FILARIOSE À TAHITI: EXAMEN DE LA SITUATION DE NOVEMBRE 1967 À JANVIER 1968

Des enquêtes sur la filariose menées à Tahiti peu de temps après la deuxième guerre mondiale ont montré le taux de prévalence le plus élevé du Pacifique sud: l'indice microfilarien atteignait 25 à 44%, la fréquence de l'éléphantiasis était de 5% et la densité microfilarienne médiane ( $DMf_{50}$ ) variait de 18 à 31.

En 1950, le district de Vairao, comptant 825 habitants a été choisi comme secteur témoin où le traitement par la diéthylcarbamazine a été retardé jusqu'en 1956. L'indice microfilarien était de 30,9% en 1950 et de 31,7% en 1956 avec des  $DMf_{50}$  de 28 et 26 respectivement. En 1960, après 4 ans de traitement de masse, l'indice microfilarien et la  $DMf_{50}$  s'étaient abaissés à 3% et 3,0.

Dans les districts de Mataiea et de Tautira, les indices microfilariens étaient de 39% et 27% en 1950 (avant tout traitement) avec des  $DMf_{50}$  de 23 et 18. En 1958, 3 à 4 ans après la fin du traitement de masse, les indices

étaient de 4,9% (Mataiea) et de 3,3% (Tautira) et les  $DMf_{50}$  de 4,5 (Mataiea) et de 4,0 (Tautira). En 1966, les valeurs s'établissaient aux taux suivants: indices microfilariens: 5,2% (Mataiea) et 4,0% (Tautira) et  $DMf_{50}$ : 3,5 (Mataiea) et 3,5 (Tautira).

Les essais de chimioprophylaxie de masse ont montré que l'administration une fois par mois d'une dose de diéthylcarbamazine de 6 mg par kilo de poids corporel pendant 12 mois (dose totale: 72 mg/kg) constituait le schéma minimal pour une lutte efficace contre la filariose. Comme mesure complémentaire, on n'a eu recours qu'à la destruction des gîtes larvaires dans un périmètre de 100 m autour de chaque habitation. La meilleure méthode pour évaluer les progrès d'un programme est probablement de comparer avant et après le traitement les indices microfilariens et les proportions de larves de *Wuchereria bancrofti* au stade infectant chez les moustiques.

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